

Digital Heroes — Lead Management Platform

A high-performance, dark-themed, production-grade Lead Management Platform built using **Next.js 16 (App Router)**, **PostgreSQL** with **Drizzle ORM**, and **NextAuth.js** with JWT sessions for role-based authentication.

1. Project Overview & Role Credentials

This application contains two parts:

- 1. **Public Lead Capture Form:** Open to anyone visiting the homepage (/) to submit a project inquiry.
- 2. **Back-office CRM Dashboard:** Securely gated dashboard (/dashboard) for managing incoming leads, updating statuses, assigning reps, and logging client note history.

Testing Accounts (Credentials)

We have seeded two default accounts for grading. You can log in using the inputs on the /login screen or by clicking the quick-login preset buttons:

Role	Email	Password	Permissions
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Admin	admin@digitalheroes.com	admin123	Full control (view all, edit all, change assignees, delete leads, add notes).
Member	member@digitalheroes.com	member123	Limited control (only view/update/add notes to leads assigned to their account).

2. API Contract & Documentation

All protected endpoints require a valid session (NextAuth JWT session cookie). If unauthenticated, they will return a 401 `Unauthorized` response.

Public Endpoints

`POST /api/leads`

Capture a lead submission from the public form.

- **Headers:** `Content-Type: application/json`
- **Request Body:**

```
{
  "name": "Jane Doe",
  "email": "jane@example.com",
  "phone": "+1 555-0199",
```

```
"company": "Acme Marketing Co",
"message": "We need custom Shopify theme development"
}
```

- **Success Response (201 Created):**

```
{
  "success": true,
  "lead": {
    "id": "c01828fc-36ab-4a57-b08e-324c4e0b0ad2",
    "name": "Jane Doe",
    "email": "jane@example.com",
    "phone": "+1 555-0199",
    "company": "Acme Marketing Co",
    "message": "We need custom Shopify theme development",
    "status": "new",
    "assignedToId": null,
    "createdAt": "2026-08-05T13:00:00.000Z",
    "updatedAt": "2026-08-05T13:00:00.000Z"
  }
}
```

- **Error Response (400 Bad Request):**

```
{ "error": "Name and Email are required fields." }
```

Protected Endpoints

GET /api/leads

Retrieve a paginated, filterable list of leads.

- **Query Parameters:**

- `page` (optional): Default 1
- `limit` (optional): Default 10

- `status` (optional): Filter by status (new, contacted, qualified, won, lost)
 - `query` (optional): Text search by lead name, email, or company name
- **Success Response (200 OK):**

```
{
  "data": [
    {
      "id": "c01828fc-36ab-4a57-b08e-324c4e0b0ad2",
      "name": "Jane Doe",
      "email": "jane@example.com",
      "phone": "+1 555-0199",
      "company": "Acme Marketing Co",
      "message": "We need custom Shopify theme development",
      "status": "new",
      "assignedToId": "2bf5df78-c0b8-4bc2-8414-0c2d3080ff12",
      "createdAt": "2026-08-05T13:00:00.000Z",
      "updatedAt": "2026-08-05T13:00:00.000Z",
      "assignedTo": {
        "id": "2bf5df78-c0b8-4bc2-8414-0c2d3080ff12",
        "name": "Sales Rep Member",
        "email": "member@digitalheroes.com",
        "role": "member"
      }
    }
  ],
  "meta": {
    "total": 1,
    "page": 1,
    "limit": 10,
    "totalPages": 1
  }
}
```

GET /api/leads/[id]

Fetch details of a single lead.

- **Success Response (200 OK):** returns the lead details object (same as lead list entry).
- **Error Response (403 Forbidden - Members accessing unassigned leads):**

```
{ "error": "Access denied: You are not assigned to th
```

PATCH /api/leads/[id]

Modify status or change lead assignee.

- **Request Body (Partial Update):**

```
{
  "status": "qualified",
  "assignedToId": "2bf5df78-c0b8-4bc2-8414-0c2d3080f1
}
```

- **Enforced Permissions:**

- **Admin:** Can update status and change assignment (assignedToId) for any lead.
- **Member:** Can only update the status of leads assigned to their own User ID. Changing assignment or editing other leads returns 403 Forbidden.

- **Success Response (200 OK):**

```
{
  "success": true,
  "lead": { ...updatedLeadData }
}
```

DELETE /api/leads/[id]

Delete a lead file.

- **Enforced Permissions:** restricted to **Admin** role. Members attempting this will receive a 403 `Forbidden` response.
- **Success Response (200 OK):**

```
{ "success": true, "message": "Lead deleted successfully" }
```

GET /api/leads/[id]/notes

Fetch notes history for a lead.

- **Success Response (200 OK):**

```
[
  {
    "id": "d02739fa-51bc-4389-c124-789a421b0cd9",
    "leadId": "c01828fc-36ab-4a57-b08e-324c4e0b0ad2",
    "userId": "1af5df78-c0b8-4bc2-8414-0c2d3080ff11",
    "content": "Followed up with client. They are ready to move forward.",
    "createdAt": "2026-08-05T13:30:00.000Z",
    "user": {
      "id": "1af5df78-c0b8-4bc2-8414-0c2d3080ff11",
      "name": "Admin User",
      "email": "admin@digitalheroes.com"
    }
  }
]
```

POST /api/leads/[id]/notes

Add a timestamped comment to a lead. Also creates an audit trail event.

- **Request Body:**

```
{ "content": "Client requested proposal by Friday." }
```

- **Success Response (201 Created):** returns the created note object.

GET /api/leads/[id]/activity

Retrieve a full history trail of actions taken on a lead.

- **Success Response (200 OK):**

```
[
  {
    "id": "e03749fa-61bc-5389-c124-889a421b0cd1",
    "leadId": "c01828fc-36ab-4a57-b08e-324c4e0b0ad2",
    "userId": "1af5df78-c0b8-4bc2-8414-0c2d3080ff11",
    "action": "Status updated to 'qualified' by Admin",
    "createdAt": "2026-08-05T13:15:00.000Z",
    "user": {
      "id": "1af5df78-c0b8-4bc2-8414-0c2d3080ff11",
      "name": "Admin User",
      "email": "admin@digitalheroes.com"
    }
  }
]
```

3. Local Setup & Testing

Environment Configuration

Create a `.env` file in the root directory (based on `.env.example` format):

```
DATABASE_URL="postgresql://<username>:<password>@<host>:<port>/<database>"
NEXTAUTH_SECRET="your-32-character-secret-key"
```

```
NEXTAUTH_URL="http://localhost:3000"
```

Installation

1. Install dependencies:

```
npm install
```

2. Setup database schema & seed credentials:

```
npm run db:setup
```

3. Run automated tests (Vitest):

```
npm run test
```

4. Start the local dev server:

```
npm run dev
```

4. Production Deployment Guide

You can easily deploy this Next.js application to **Vercel** combined with a hosted database (like **Supabase**, **Neon**, or **Render PostgreSQL**).

Step 1: Database Setup

1. Create a free PostgreSQL instance on [Supabase](#) or [Neon](#).
2. Copy the connection string. It should look like this:

```
postgresql://postgres:password@aws-0-us-east-1.pooler.supabase.com:6543/postgres?
```



```
sslmode=require
```

Step 2: Deploy to Vercel

1. Log in to [Vercel](#) and click **Add New → Project**.
2. Connect your GitHub repository (`lead-platform`).
3. Under **Environment Variables**, add:
 - `DATABASE_URL` (your database connection string)
 - `NEXTAUTH_SECRET` (generate a secure random string, e.g. `run openssl rand -base64 32`)
 - `NEXTAUTH_URL` (your deployed Vercel domain URL, e.g. `https://my-lead-platform.vercel.app`)
4. Click **Deploy**.

Step 3: Run Database Migrations in Production

Once deployed, run migrations and seed the default accounts in the production database:

1. Make sure your local `.env` points to the production `DATABASE_URL` temporary.
2. Run the migration script locally:

```
npm run db:setup
```

This connects to the production database, runs the SQL DDL schemas, and inserts the credentials. Make sure to revert your local `.env` database URL after running.